THE INTERNAL HOM INTO AN INTERNAL HOM IS THE INTERNAL HOM FROM THE DERIVED TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of representatives, the displayed identity of complexes from the composition lemma, the identification of the two sides as derived internal homs, and the recovery of the chapter's global-sections equation.

Editorial note. The label lemma-internal-hom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. This card belongs to the priority queue of the D-72 twin ruling (the sites-side statement is the designated counting side, now counted here for the first time). The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Composition of internal homs. Same chapter, the lemma identifying the complex of internal homs into an internal hom with the internal hom from the total tensor product. Supplies the displayed identity of complexes at the heart of the proof.
- The internal hom from a K-flat complex into a K-injective complex is K-injective. Same chapter, the lemma stating that the internal hom from a K-flat complex into a K-injective complex is K-injective. Supplies that the left-hand complex represents the derived internal hom into the derived internal hom.
- Zeroth cohomology of the derived internal hom as derived Hom. Same chapter, the lemma identifying H^0 of the internal hom complex with the derived Hom groups over an object and globally. Supplies the identification of the global sections formula after taking H^0.

Source reference. Original equation 08DI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8844–8844. Prerequisite pointer invoked at proof line(s) 8905; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A8T: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8789–8789. Prerequisite pointer invoked at proof

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line(s) 8925; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A5Y: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8616–8616. Prerequisite pointer invoked at proof line(s) 8922; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08DK: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8874–8874. Prerequisite pointer invoked at proof line(s) 8929; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let K, L, M be objects of $D(\mathcal{O})$. With the construction as described above there is a canonical isomorphism*

$$R\mathcal{H}om(K, R\mathcal{H}om(L, M)) = R\mathcal{H}om(K \otimes^{\mathbf{L}}_{\mathcal{O}} L, M)$$

in $D(\mathcal{O})$ functorial in K, L, M which recovers (08DI) on taking $H^0(\mathcal{C}, -)$.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing M and a K-flat complex of $\mathcal{O}$-modules $\mathcal{L}^\bullet$ representing L. For any complex of $\mathcal{O}$-modules $\mathcal{K}^\bullet$ we have

$$\mathcal{H}om^\bullet(\mathcal{K}^\bullet, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)) = \mathcal{H}om^\bullet(\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet), \mathcal{I}^\bullet)$$

by Lemma 0A5Y. Note that the left hand side represents $R\mathcal{H}om(K, R\mathcal{H}om(L, M))$ (use Lemma 0A8T) and that the right hand side represents $R\mathcal{H}om(K \otimes^{\mathbf{L}}_{\mathcal{O}} L, M)$. This proves the displayed formula of the lemma. Taking global sections and using Lemma 08DK we obtain (08DI). □